

# Victor Wong

Boston, MA | victorwong315@gmail.com | linkedin.com/in/victorw12/ | github.com/VictorWong123 | 617-797-9352

## EDUCATION

### Union College

Schenectady, NY

*Bachelor of Science in Computer Science*

*Expected Graduation: June 2028*

- **GPA:** 3.72 | Honors: Dean's List
- **Coursework:** Programming with Purpose, Data Structures, Statistics, Web Development, Data Visualization
- **Activities:** International Collegiate Programming Contest (ICPC), Codepath, UR2Phd, MLT Career Prep Fellow

## SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, HTML, CSS

**Technologies:** React, Express.js, NodeJS, Tailwind, D3.js, MongoDB, SQL, PostgreSQL, Git, Docker, FastAPI

## WORK EXPERIENCE

### Engineering Intern

Jun 2026 – Present

*The First Electronics Corporation*

*Braintree, MA*

- Reduced manual data entry time by building an OCR pipeline with Tesseract.js and Python to convert legacy paper records into structured Excel data.

### Undergraduate HCI Researcher

Oct 2024 – Present

*Union College*

*Schenectady, NY*

- Programmed a real-time web app with MediaPipe to detect 5 distinct gestures, enabling hands-free interaction in a large-screen Would You Rather game.
- Ran a 50+ participant study, advancing understanding of social interaction with large screens after reviewing 30+ Human-Computer Interaction and UI/UX papers.
- Launched CAP Lab's first website using Next.js, increasing visibility of HCI research to 2,000+ students.

### Software Engineering Intern

Apr 2026 – Jun 2026

*BIS Computer Solutions*

*Remote*

- Created a natural language-to-SQL engine using Gemini, TypeScript, and vector search, enabling non-technical users to securely query inventory data.
- Automated support intake by building an AI voice agent that answers inbound calls and creates structured Jira tickets, reducing manual handoffs and speeding issue routing.

### Software Engineering Intern

Feb 2025 – Jun 2025

*EpiBuild*

*Remote*

- Developed 3 interactive D3.js visualizations to visualize 20,000+ sales records, giving managers real-time insights.
- Optimized SQL data processing pipelines to clean and aggregate 20,000+ sales records, reducing analysis time by 20%.
- Managed cloud deployment on Heroku and AWS S3 to process and enable querying of 2GB of historical sales data, ensuring fast retrieval.

## PROJECTS

### BioRT-Bench (2nd Place AIBio Hackathon: Biosecurity Track) | *Python, PyRIT, React, TypeScript, FastAPI*

- Built an LLM red-teaming benchmark evaluating 4 frontier models across 4 jailbreak attacks and 40 biosecurity prompts, measuring safeguard robustness under adversarial pressure.
- Developed a StrongREJECT-inspired scoring system and React dashboard to analyze refusal, specificity, and actionability across 640 model responses.

### Viva (NexHacks Project) | *FastAPI, React, TypeScript, Python LiveKit, MongoDB*

- Built a real-time voice tutoring system using LiveKit WebRTC and FastAPI, achieving <3s audio latency.
- Integrated Google Classroom APIs with OAuth and MongoDB token storage to securely fetch courses and assignments.
- Personalized the tutoring experience by using Gemini to replicate teachers' teaching styles from audio while summarizing student challenges for teachers.

### FlexFlow | *React, TypeScript, Python, FastAPI, LiveKit, Gemini, Supabase, Next.js*

- Architected a real-time AI physical therapist with sub-100ms latency, using a ThreadPoolExecutor to process MediaPipe landmarks without blocking the async audio stream.
- Deployed a weighted search engine to retrieve images for 873+ exercises, applying synonym expansion for precise form.
- Automated session tracking by using Gemini to summarize sessions, creating a history of the user's progress.